Practical 09

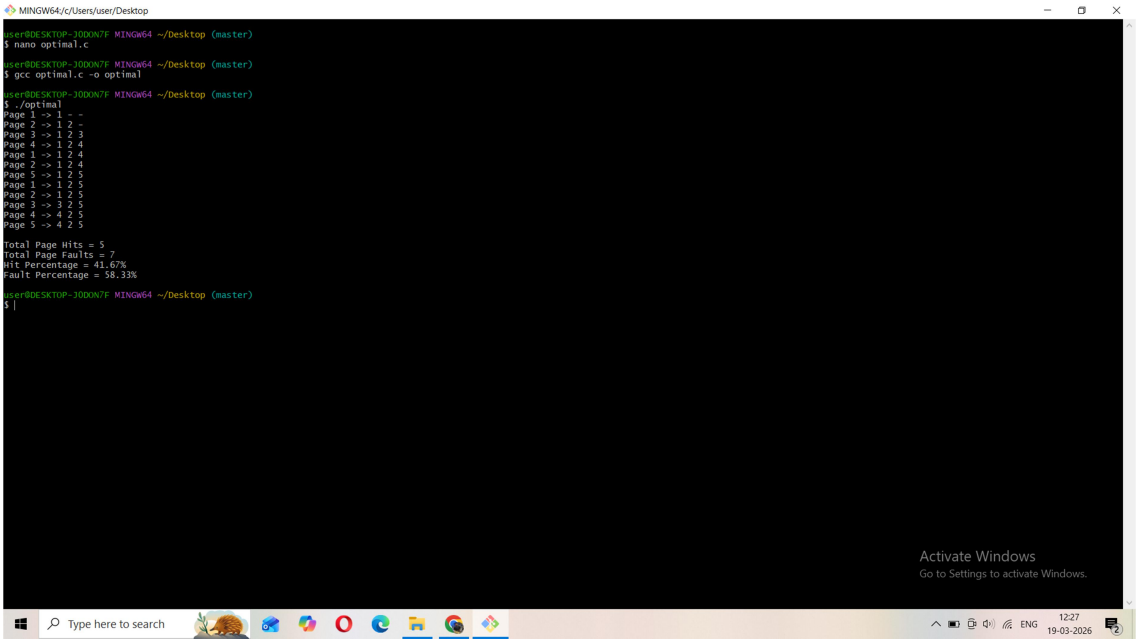
Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:
```
#include <stdio.h>

int main() {
   int frames = 3;
   int pages[] = {1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5};
   int n = sizeof(pages) / sizeof(pages[0]);

   int frame[10], hit = 0, fault = 0;

   for (int i = 0; i < frames; i++)
      frame[i] = -1;

   for (int i = 0; i < n; i++) {
      int found = 0;

      for (int j = 0; j < frames; j++) {
         if (frame[j] == pages[i]) {
            hit++;
            found = 1;
            break;
         }
```

```
}

if (!found) {
    int index = -1, farthest = i + 1;

    for (int j = 0; j < frames; j++) {
        int k;
        for (k = i + 1; k < n; k++) {
            if (frame[j] == pages[k]) {
                if (k > farthest) {
                    farthest = k;
                    index = j;
                }
                break;
            }
        }

        if (k == n) {
            index = j;
            break;
        }
    }

    if (index == -1)
        index = 0;

    frame[index] = pages[i];
```

```
            fault++;
        }

        printf("Page %d -> ", pages[i]);
        for (int j = 0; j < frames; j++) {
            if (frame[j] != -1)
                printf("%d ", frame[j]);
            else
                printf("- ");
        }
        printf("\n");
    }

    float hit_ratio = (float)hit / n * 100;
    float fault_ratio = (float)fault / n * 100;

    printf("\nTotal Page Hits = %d", hit);
    printf("\nTotal Page Faults = %d", fault);
    printf("\nHit Percentage = %.2f%%", hit_ratio);
    printf("\nFault Percentage = %.2f%%\n", fault_ratio);

    return 0;
}
```

Output:



```
MINGW64:/c/Users/user/Desktop

user@DESKTOP-JODON7F MINGW64 ~/Desktop (master)
$ nano optimal.c

user@DESKTOP-JODON7F MINGW64 ~/Desktop (master)
$ gcc optimal.c -o optimal

user@DESKTOP-JODON7F MINGW64 ~/Desktop (master)
$ ./optimal
Page 1 -> 1 - -
Page 2 -> 1 2 -
Page 3 -> 1 2 3
Page 4 -> 1 2 4
Page 1 -> 1 2 4
Page 2 -> 1 2 4
Page 5 -> 1 2 5
Page 1 -> 1 2 5
Page 2 -> 1 2 5
Page 3 -> 3 2 5
Page 4 -> 4 2 5
Page 5 -> 4 2 5

Total Page Hits = 5
Total Page Faults = 7
Hit Percentage = 41.67%
Fault Percentage = 58.33%

user@DESKTOP-JODON7F MINGW64 ~/Desktop (master)
$ |
```

Activate Windows
Go to Settings to activate Windows.

Type here to search
ENG
12:27
19-03-2026